

MediCamp

A Community Healthcare Camp Management System

Prepared for

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1. Project Title

MediCamp – A Community Healthcare Camp Management System

2. Problem Statement

Free medical camps run by NGOs are one of the main ways rural and underserved communities in Bangladesh access healthcare. Most camps still run on paper: hand-written registration forms, paper prescriptions, and manual stock counts.

This causes concrete problems. Patients lose continuity of care because doctors cannot see prior diagnoses or allergies from earlier camps. Pharmacists run out of stock without warning or waste unused medicine because counts are done by hand. Referrals and follow-ups are rarely tracked end-to-end, so patients whose cases need hospital care can fall through the cracks. Area-wise reports for donors and organisers take days to compile manually, which slows funding decisions and future camp planning.

2.1 Aligned Sustainable Development Goal

Of the 17 UN Sustainable Development Goals, MediCamp's proposal maps most directly onto:

- SDG 3 – Good Health and Well-being: the project's core purpose is to make free community health camps more reliable and effective, from registration and diagnosis through prescription, referral, and follow-up, directly improving access to essential healthcare for underserved populations.

The system also has secondary alignment with SDG 10 (Reduced Inequalities), since it targets rural and low-income communities that lack regular access to care, but SDG 3 is the primary goal the proposal addresses.

3. Project Description & Solution Approach

MediCamp is a role-based web application covering the full life cycle of a community healthcare camp. It supports six actors — Admin, Host (organising NGO), Doctor, Volunteer, Pharmacist, and Patient — each with a dashboard scoped only to the actions their role needs.

The system models a camp as a workflow that runs from planning and staffing, through patient registration and screening, consultation, prescription and dispensing, to referral, follow-up, and reporting. A Blood Donation module and a multi-level (district-to-village) Reporting module sit on top of the same shared patient and camp data, so information entered once at registration is reused everywhere it is needed.

The solution approach favours a normalised relational database (PostgreSQL) with foreign keys, transactions, and triggers over a loosely structured store, since the domain is highly relational — patients link to visits, visits to diagnoses and prescriptions, prescriptions to inventory, camps to staff and budgets — and a change in one place must stay consistent everywhere else. The database itself enforces integrity, for example ensuring medicine stock never goes negative. On top of this, an ASP.NET Core MVC application layer provides role-based authentication and authorisation, so each of the six actors only sees the screens relevant to their job.

4. Key Functionalities / Features

4.1 Admin

- Create and manage user accounts; assign roles and permissions
- Approve or reject Host (NGO) registration requests
- Maintain the location hierarchy (district, upazila, union, village)

- Maintain master reference lists (medicines, hospitals, blood groups)
- Generate system-wide, cross-Host reports

4.2 Host (Organising NGO)

- Create and configure a camp (name, type, location, schedule, budget)
- Assign doctors, volunteers, and pharmacists to a camp
- Allocate medicine and equipment inventory from available stock
- Publish the camp and track expenses against budget

4.3 Doctor

- Review a patient's visit history, diagnoses, and allergies
- Record diagnosis, clinical notes, and advice for a visit
- Create a prescription that is sent directly to the pharmacy
- Refer a patient to a hospital or flag a required follow-up

4.4 Volunteer

- Register a new patient or search for an existing one
- Generate a queue token and place the patient in the waiting queue
- Record vitals, BMI, blood pressure, and presenting symptoms
- Conduct follow-up visits and record recovery status

4.5 Pharmacist

- View incoming prescriptions from the consultation phase
- Dispense medicine, which automatically decreases camp inventory
- Mark a medicine out-of-stock and submit a restock request

4.6 Patient

- Register an account and view upcoming camp schedules
- Register for a camp and view personal visit/prescription history
- Opt in as a blood donor, search donors, or submit a blood request

5. System Diagrams

The diagram below shows MediCamp's layered system architecture: role-scoped presentation views on top, an ASP.NET Core MVC application layer enforcing role-based access, the core domain modules, a data access layer via Entity Framework Core, and the underlying PostgreSQL database, with the deployment targets used for hosting.

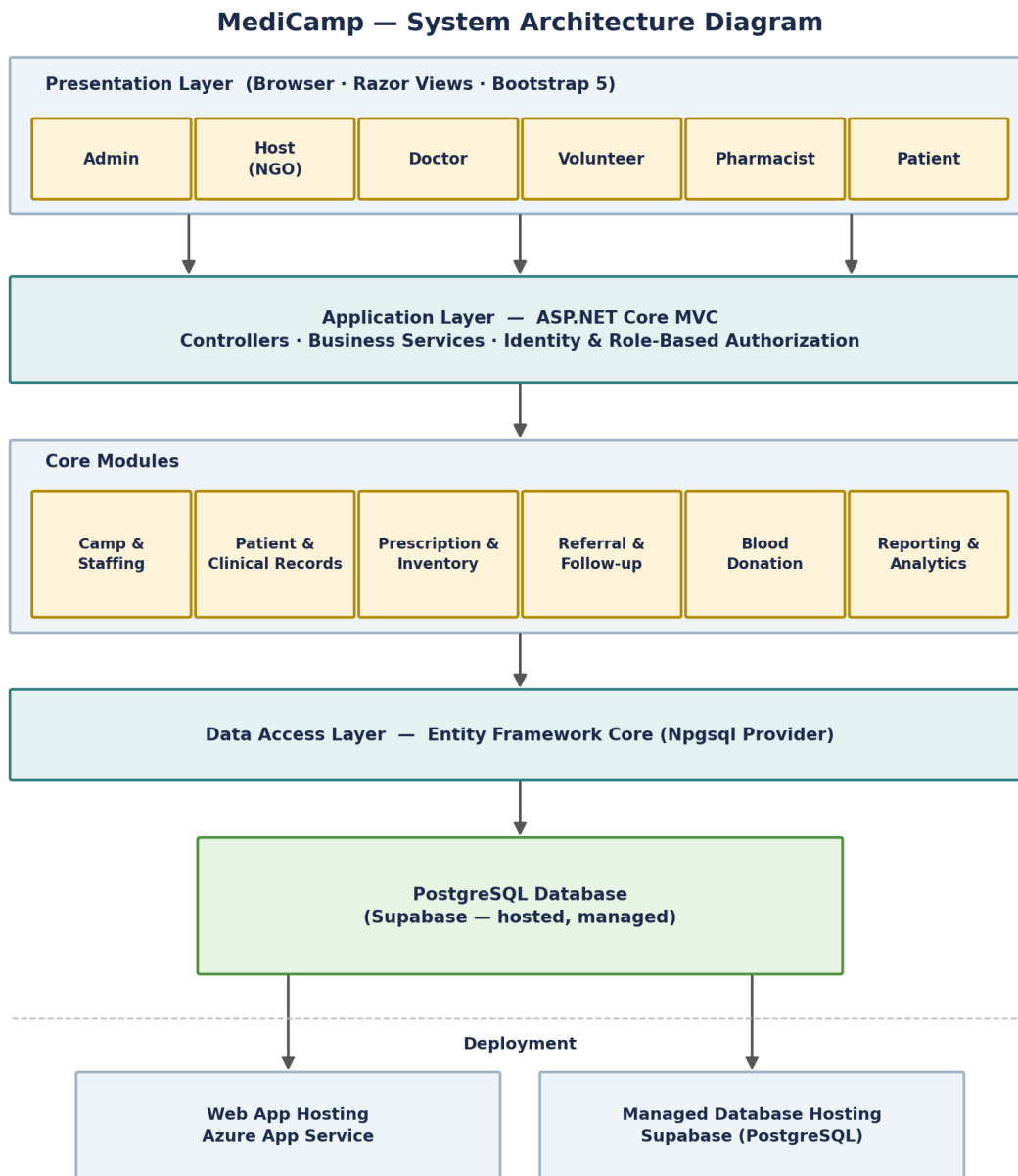


Figure 1: System Architecture Diagram — MediCamp

5.1 Entity–Relationship Diagram

The ER diagram models an 18-entity relational schema covering location hierarchy, camp organisation, staffing, patient visits, clinical records, prescriptions/inventory, referrals, and the blood donation workflow.

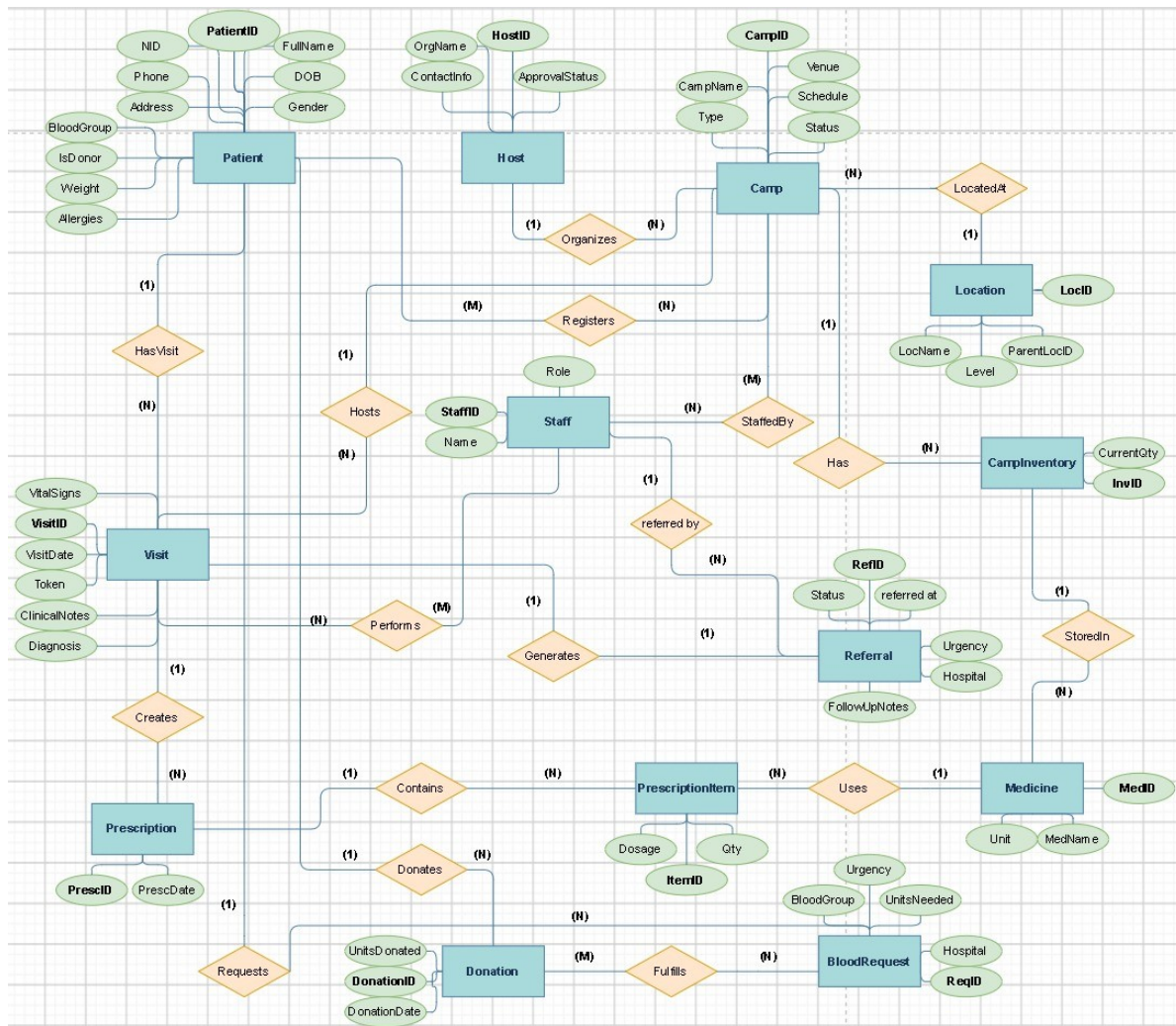


Figure 2: Entity-Relationship Diagram — MediCamp

6. Background Study / Gap Analysis

Free health camps are a long-standing model used by charities and NGOs across Bangladesh: short-duration, staffed largely by volunteer doctors, and run on a fixed donation budget — very different from a permanent hospital's continuous staffing and IT support.

6.1 Existing Systems Considered

- General-purpose Hospital Information / EMR systems (e.g. OpenMRS, Bahmni): built for permanent facilities with continuous connectivity and dedicated IT staff.
- Spreadsheet / paper-based tracking: the current default for most NGO-run camps in Bangladesh.
- Tele-consultation platforms: Bangladesh's digital-health efforts have mostly focused on remote consultation and permanent hospital records, not on field camps.

6.2 Gaps Identified

Aspect	Existing Systems	MediCamp
Event model	Assume a permanent, continuously staffed facility	Models a camp as a short, budgeted, staffed event
Patient continuity	Records isolated per visit or per facility	One longitudinal patient record reused across camps
Inventory	Manual or hospital-scale stock systems	Camp-scoped inventory with auto-deduction and shortage flags
Connectivity assumptions	Assume stable, high-speed internet	Designed to tolerate weak/rural connectivity
Blood donation	Rarely included, or a separate system	Integrated donor registry and request workflow
Reporting	Facility-level, not area-based	Area-wise (district to village) reports for donors/Hosts

This confirms a gap: a purpose-built, camp-centric system that treats each camp as a budgeted, temporarily staffed event — while still keeping one longitudinal patient record and enabling area-wise reporting — is missing from both general hospital software and the ad-hoc paper/spreadsheet processes NGOs currently rely on. MediCamp is designed specifically to close this gap.